

23/10/2025

What we tightened up

- **Attach as soon as it's possible.** Right after `GlTFLoader` finishes, we try an immediate attach in case the reference image is already in view. Fewer “why isn't it showing yet?” moments.
- **No more floating meshes.** The model stays hidden until we've parented it under the tracked page and scaled to the page width.
- **Clearer UI guidance.** Two small prompts that do real work:
 - “Point at the **same picture** you snapped.”
 - “Fill ~40–60% of the screen and avoid glare.”

How the scripts collaborate

- **GenerateController:** the conductor. Wires Camera → RuntimeRef → Meshy → GLB → Overlay. Handles cancel/retry and remembers the last reference name + model.
- **CameraCapture:** one solid photo, two outputs (JPG bytes + Texture2D).
- **RuntimeImageLibraryController:** a mutable library with a real-world width (we default ~0.15 m). Stores `CurrentRefName`. Can show a green debug quad when tracking hits.
- **MeshyClient:** create job, poll status, return GLB URL.
- **GlTFLoader:** download + load GLB with progress and error callbacks.
- **ModelOverlayController:** listens to `trackedImagesChanged`, filters by `CurrentRefName` + `Tracking`, then parents + scales + offsets, and can lock to the first good hit.
- **UIController:** progress, errors, Cancel/Retry, and “Model ready ✓”.

Field notes we now believe

- **Lighting matters** more than we wanted it to. We say it plainly in the UI now.
- **Bigger is kinder.** If the page image is tiny in frame, tracking is flaky. Filling about half the screen is a sweet spot.
- **Device quirks are real.** Our “runtime add nudge” is now a standard utility we flip on when needed.

What's next

- A tiny **AR card** in the UI showing the last reference snapshot, so people know exactly what to aim at again.
- Optional **tap-to-place** fallback for weak images (so you can still play, just without image anchoring).
- Local **GLB caching** so previously generated models can work offline.
- **Batch generation** and a lightweight gallery to manage multiple characters.

Quick cheat-sheet we keep nearby

- **Flow:** Generate → photo → add to runtime library → Meshy job → poll → download GLB → load → point at the **same** picture → model appears.

23/10/2025

- **If nothing shows:** confirm the camera sees the original picture; look for `state=Tracking` in logs; try a higher-contrast image; rebind the image manager if runtime add didn't take.